

Now,

$$\begin{aligned}
 (m+1)(m+1+1)[2(m+1)+1] &= (m+1)(m+2)\{(2m+1)+2\} \\
 &= (m+1)(m+2)(2m+1)+2(m+1)(m+2) \\
 &= m(m+1)(2m+1)+2(m+1)(2m+1)+2(m+1)(m+2) \\
 &= m(m+1)(2m+1)+2(m+1)(2m+1+m+2) \\
 &= m(m+1)(2m+1)+2(m+1)(3m+3) \\
 &= m(m+1)(2m+1)+6(m+1)^2 = 6\lambda+6(m+1)^2 \quad [\text{Using (i)}] \\
 &= 6\{\lambda+(m+1)^2\}, \text{ which is divisible by 6}
 \end{aligned}$$

$\Rightarrow P(m+1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true

Hence, by the principle of mathematical induction, the given statement is true for all $n \in N$.

EXAMPLE 3 Prove by the principle of mathematical induction that for all $n \in N$:

$$1+4+7+\dots+(3n-2) = \frac{1}{2}n(3n-1)$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n): 1+4+7+\dots+(3n-2) = \frac{1}{2}n(3n-1)$$

STEP I We have,

$$P(1): 1 = \frac{1}{2} \times (1) \times (3 \times 1 - 1).$$

$$\therefore 1 = \frac{1}{2} \times (1) \times (3 \times 1 - 1)$$

So, $P(1)$ is true

STEP II Let $P(m)$ be true. Then,

$$1+4+7+\dots+(3m-2) = \frac{1}{2}m(3m-1) \quad \dots(i)$$

We wish to show that $P(m+1)$ is true. For this we have to show that

$$1+4+7+\dots+(3m-2)+\{3(m+1)-2\} = \frac{1}{2}(m+1)\{3(m+1)-1\}$$

Now, $1+4+7+\dots+(3m-2)+\{3(m+1)-2\}$

$$= \frac{1}{2}m(3m-1)+\{3(m+1)-2\} \quad [\text{Using (i)}]$$

$$= \frac{1}{2}m(3m-1)+(3m+1) = \frac{1}{2}\{3m^2-m+6m+2\}$$

$$= \frac{1}{2}\{3m^2+5m+2\} = \frac{1}{2}(m+1)(3m+2) = \frac{1}{2}(m+1)\{3(m+1)-1\}$$

$\therefore P(m+1)$ is true

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction, the given result is true for all $n \in N$.

EXAMPLE 4 Prove by the principle of mathematical induction that for all $n \in N$:

$$1^2+2^2+3^2+\dots+n^2 = \frac{1}{6}n(n+1)(2n+1)$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n): 1^2+2^2+3^2+\dots+n^2 = \frac{1}{6}n(n+1)(2n+1)$$

STEP I We have,

$$P(1): 1^2 = \frac{1}{6}(1)(1+1)(2 \times 1 + 1)$$

$$\therefore 1^2 = 1 = \frac{1}{6} (1) (1+1) (2 \times 1 + 1)$$

$\therefore P(1)$ is true

STEP II Let $P(m)$ be true. Then,

$$1^2 + 2^2 + 3^2 + \dots + m^2 = \frac{1}{6} m(m+1)(2m+1) \quad \dots(i)$$

We wish to show that $P(m+1)$ is true. For this we have to show that

$$1^2 + 2^2 + 3^2 + \dots + m^2 + (m+1)^2 = \frac{1}{6} (m+1) \{(m+1)+1\} \{2(m+1)+1\}$$

$$\text{Now, } 1^2 + 2^2 + 3^2 + \dots + m^2 + (m+1)^2$$

$$= \{1^2 + 2^2 + 3^2 + \dots + m^2\} + (m+1)^2$$

$$= \frac{1}{6} m(m+1)(2m+1) + (m+1)^2 \quad [\text{Using (i)}]$$

$$= \frac{1}{6} (m+1) \{m(2m+1) + 6(m+1)\} = \frac{1}{6} (m+1) \{2m^2 + 7m + 6\}$$

$$= \frac{1}{6} (m+1)(m+2)(2m+3) = \frac{1}{6} (m+1) \{(m+1)+1\} \{2(m+1)+1\}$$

$\therefore P(m+1)$ is true

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction, the given result is true for all $n \in N$.

EXAMPLE 5 Using the principle of mathematical induction prove that:

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left\{ \frac{n(n+1)}{2} \right\}^2 \text{ for all } n \in N$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n): 1^3 + 2^3 + 3^3 + \dots + n^3 = \left\{ \frac{n(n+1)}{2} \right\}^2$$

STEP I We have,

$$P(1): 1^3 = \left\{ \frac{1(1+1)}{2} \right\}^2$$

$$\text{Clearly, } 1^3 = 1 = \left\{ \frac{1(1+1)}{2} \right\}^2$$

$\therefore P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$1^3 + 2^3 + 3^3 + \dots + m^3 = \left\{ \frac{m(m+1)}{2} \right\}^2 \quad \dots(ii)$$

We shall now prove that $P(m+1)$ is true. For this we have to prove that

$$1^3 + 2^3 + 3^3 + \dots + m^3 + (m+1)^3 = \left\{ \frac{(m+1)\{(m+1)+1\}}{2} \right\}^2$$

Now,

$$1^3 + 2^3 + 3^3 + \dots + m^3 + (m+1)^3$$

$$= \{1^3 + 2^3 + \dots + m^3\} + (m+1)^3$$

$$= \left\{ \frac{m(m+1)}{2} \right\}^2 + (m+1)^3 \quad [\text{Using (i)}]$$

$$\begin{aligned}
 &= (m+1)^2 \left\{ \frac{m^2}{4} + (m+1) \right\} \\
 &= (m+1)^2 \left\{ \frac{m^2 + 4m + 4}{4} \right\} = \frac{(m+1)^2 (m+2)^2}{4} = \left\{ \frac{(m+1) [(m+1) + 1]}{2} \right\}^2
 \end{aligned}$$

$\therefore P(m+1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction, the given result is true for all $n \in N$.

EXAMPLE 6 Using the principle of mathematical induction, prove that

$$1.2.3 + 2.3.4 + \dots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4} \text{ for all } n \in N.$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : 1.2.3 + 2.3.4 + \dots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

STEP I We have,

$$P(1) : 1.2.3 = \frac{1(1+1)(1+2)(1+3)}{4}$$

$$\therefore 1.2.3 = 6 \text{ and } \frac{1(1+1)(1+2)(1+3)}{4} = \frac{2 \times 3 \times 4}{4} = 6$$

$$\therefore 1.2.3 = \frac{1(1+1)(1+2)(1+3)}{4}$$

So, $P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$1.2.3 + 2.3.4 + \dots + m(m+1)(m+2) = \frac{m(m+1)(m+2)(m+3)}{4} \quad \dots(i)$$

We shall now show that $P(m+1)$ is true. For this we will prove that

$$\begin{aligned}
 1.2.3 + 2.3.4 + \dots + m(m+1)(m+2) + (m+1)(m+2)(m+3) \\
 = \frac{(m+1)(m+2)(m+3)(m+4)}{4}
 \end{aligned}$$

Now, $1.2.3 + 2.3.4 + \dots + m(m+1)(m+2) + (m+1)(m+2)(m+3)$

$$= \frac{m(m+1)(m+2)(m+3)}{4} + (m+1)(m+2)(m+3)$$

[Using (i)]

$$= (m+1)(m+2)(m+3) \left(\frac{m}{4} + 1 \right) = \frac{(m+1)(m+2)(m+3)(m+4)}{4}$$

$\therefore P(m+1)$ is true.

Hence, by the principle of mathematical induction $P(n)$ is true for all $n \in N$.

EXAMPLE 7 Using the principle of mathematical induction prove that

$$1.3 + 2.3^2 + 3.3^3 + \dots + n.3^n = \frac{(2n-1)3^{n+1} + 3}{4} \text{ for all } n \in N$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : 1.3 + 2.3^2 + 3.3^3 + \dots + n.3^n = \frac{(2n-1)3^{n+1} + 3}{4}$$

$$\text{STEP I } P(1) : 1.3 = \frac{(2 \times 1 - 1) \times 3^{1+1} + 3}{4}$$

$$\therefore 1.3 = 3 \text{ and } \frac{(2 \times 1 - 1) \times 3^{1+1} + 3}{4} = \frac{9 + 3}{4} = 3$$

$$\therefore 1.3 = \frac{(2 \times 1 - 1) \times 3^{1+1} + 3}{4}$$

So, $P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$1.3 + 2.3^2 + 3.3^3 + \dots + m.3^m = \frac{(2m-1)3^{m+1} + 3}{4} \quad \dots(i)$$

We shall now show that $P(m+1)$ is true.

$$\text{i.e. } 1.3 + 2.3^2 + 3.3^3 + \dots + m.3^m + (m+1).3^{m+1} = \frac{[2(m+1) - 1] 3^{(m+1)+1} + 3}{4}$$

Now,

$$\begin{aligned} & 1.3 + 2.3^2 + 3.3^3 + \dots + m.3^m + (m+1).3^{m+1} \\ &= \frac{(2m-1)3^{m+1} + 3}{4} + (m+1)3^{m+1} \\ &= \frac{(2m-1)3^{m+1} + 3 + (4m+4)3^{m+1}}{4} \\ &= \frac{(2m-1) \times 3^{m+1} + (4m+4) \times 3^{m+1} + 3}{4} \\ &= \frac{(2m-1+4m+4)3^{m+1} + 3}{4} \\ &= \frac{(6m+3)3^{m+1} + 3}{4} = \frac{(2m+1)3^{m+2} + 3}{4} = \frac{[2(m+1) - 1] 3^{(m+1)+1} + 3}{4} \end{aligned}$$

$\therefore P(m+1)$ is true.

Hence, by the principal of mathematical induction $P(n)$ is true for all $n \in N$ i.e., the given result is true for all $n \in N$.

EXAMPLE 8 Prove by the principle of mathematical induction that for all $n \in N$:

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n): \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

STEP I We have, $P(1): \frac{1}{1.2} = \frac{1}{1+1}$

$$\therefore \frac{1}{1.2} = \frac{1}{1+1} = \frac{1}{2}$$

So, $P(1)$ is true

STEP II Let $P(m)$ be true. Then,

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{m(m+1)} = \frac{m}{m+1} \quad \dots(i)$$

We shall now show that $P(m+1)$ is true. For this we have to show that

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{m(m+1)} + \frac{1}{(m+1)(m+1+1)} = \frac{(m+1)}{(m+1)+1}$$

$$\text{Now, } \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{m(m+1)} + \frac{1}{(m+1)((m+1)+1)}$$

$$\begin{aligned}
 &= \left\{ \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{m(m+1)} \right\} + \frac{1}{(m+1)((m+1)+1)} \\
 &= \frac{m}{m+1} + \frac{1}{(m+1)((m+1)+1)} = \frac{m}{m+1} + \frac{1}{(m+1)(m+2)} \quad [\text{Using (i)}] \\
 &= \frac{1}{(m+1)} \left\{ \frac{m}{1} + \frac{1}{m+2} \right\} = \frac{1}{(m+1)} \times \frac{(m^2 + 2m + 1)}{(m+2)} = \frac{(m+1)^2}{(m+1)(m+2)} \\
 &= \frac{m+1}{m+2} = \frac{(m+1)}{(m+1)+1}
 \end{aligned}$$

$\therefore P(m+1)$ is true

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true

Hence, by the principle of mathematical induction, the given statement is true for all $n \in N$.

EXAMPLE 9 Using the principle of mathematical induction prove that

$$1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \frac{1}{1+2+3+4} + \dots + \frac{1}{1+2+3+\dots+n} = \frac{2n}{n+1} \text{ for all } n \in N.$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : 1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+n} = \frac{2n}{n+1}$$

STEP I We have, $P(1) : 1 = \frac{2 \times 1}{1+1}$

$$\text{Clearly, } \frac{2 \times 1}{1+1} = \frac{2}{2} = 1$$

$$\therefore 1 = \frac{2 \times 1}{1+1}$$

So, $P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+m} = \frac{2m}{m+1} \quad \dots(i)$$

We shall now show that $P(m+1)$ is true. For this we will prove that

$$1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+m} + \frac{1}{1+2+3+\dots+(m+1)} = \frac{2(m+1)}{(m+1)+1}$$

$$\text{Now, } 1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+m} + \frac{1}{1+2+3+\dots+(m+1)}$$

$$= \frac{2m}{m+1} + \frac{1}{1+2+3+\dots+(m+1)} \quad [\text{Using (i)}]$$

$$= \frac{2m}{m+1} + \frac{1}{\frac{(m+1)(m+2)}{2}} \quad \left[\because 1+2+\dots+m+(m+1) = \frac{(m+1)(m+2)}{2} \right]$$

$$= \frac{2m}{m+1} + \frac{2}{(m+1)(m+2)} \quad [\text{Using (i)}]$$

$$= \frac{2}{m+1} \left\{ m + \frac{1}{(m+2)} \right\} = \frac{2}{m+1} \left\{ \frac{m^2 + 2m + 1}{(m+2)} \right\} = \frac{2}{m+1} \times \frac{(m+1)^2}{m+2} = \frac{2(m+1)}{(m+1)+1}$$

$\therefore P(m+1)$ is true.

Hence, by the principle of of mathematical induction $P(n)$ is true for all $n \in N$.

EXAMPLE 10 Prove by induction that the sum $S_n = n^3 + 3n^2 + 5n + 3$ is divisible by 3 for all $n \in N$.

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : S_n = n^3 + 3n^2 + 5n + 3 \text{ is divisible by 3}$$

STEP I We have, $P(1) : S_1 = 1^3 + 3(1)^2 + 5(1) + 3$ is divisible by 3.

$$\text{Since, } 1^3 + 3(1)^2 + 5(1) + 3 = 12, \text{ which is divisible by 3}$$

$$\therefore P(1) \text{ is true}$$

STEP II Let $P(m)$ be true. Then,

$$S_m = m^3 + 3m^2 + 5m + 3 \text{ is divisible by 3}$$

$$\Rightarrow S_m = m^3 + 3m^2 + 5m + 3 = 3\lambda, \text{ for some } \lambda \in N \quad \dots(i)$$

We now wish to show that $P(m+1)$ is true. For this we have to show that $(m+1)^3 + 3(m+1)^2 + 5(m+1) + 3$ is divisible by 3.

$$\text{Now, } (m+1)^3 + 3(m+1)^2 + 5(m+1) + 3$$

$$= (m^3 + 3m^2 + 5m + 3) + 3m^2 + 9m + 9$$

$$= 3\lambda + 3(m^2 + 3m + 3)$$

$$= 3(\lambda + m^2 + 3m + 3)$$

[Using (i)]

$$= 3\mu, \text{ where } \mu = \lambda + m^2 + 3m + 3 \in N$$

$$\therefore P(m+1) \text{ is true}$$

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true

Hence, by the principle of mathematical induction the statement is true for all $n \in N$.

EXAMPLE 11 Prove by the principle of mathematical induction that for all $n \in N$:

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : \frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

$$\text{STEP I} \quad \text{We have, } P(1) : \frac{1}{1.3} = \frac{1}{(2 \times 1 + 1)}.$$

$$\text{Clearly, } \frac{1}{1.3} = \frac{1}{(2 \times 1 + 1)}.$$

So, $P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2m-1)(2m+1)} = \frac{m}{2m+1} \quad \dots(i)$$

We shall now show that $P(m+1)$ is true. For this we shall show that

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2m-1)(2m+1)} + \frac{1}{(2m+1)(2m+3)} = \frac{m+1}{2m+3}$$

Now,

$$\begin{aligned} & \frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2m-1)(2m+1)} + \frac{1}{(2m+1)(2m+3)} \\ &= \frac{m}{2m+1} + \frac{1}{(2m+1)(2m+3)} \end{aligned}$$

[Using (i)]

$$= \frac{2m^2 + 3m + 1}{(2m + 1)(2m + 3)} = \frac{(2m + 1)(m + 1)}{(2m + 1)(2m + 3)} = \frac{m + 1}{2m + 3}$$

$\therefore P(m + 1)$ is true

Thus, $P(m)$ is true $\Rightarrow P(m + 1)$ is true

Hence, by the principle of mathematical induction, the given result is true for all $n \in N$.

EXAMPLE 12 Using the principle of mathematical induction, prove that

$$\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)} \text{ for all } n \in N.$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : \frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

STEP I We have,

$$P(1) = \frac{1}{1.2.3} = \frac{1(1+3)}{4(1+1)(1+2)}$$

$$\therefore \frac{1}{1.2.3} = \frac{1}{6} \text{ and } \frac{1(1+3)}{4(1+1)(1+2)} = \frac{4}{4 \times 2 \times 3} = \frac{1}{6}$$

$$\therefore \frac{1}{1.2.3} = \frac{1(1+3)}{4(1+1)(1+2)}$$

So, $P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots + \frac{1}{m(m+1)(m+2)} = \frac{m(m+3)}{4(m+1)(m+2)} \quad \dots(i)$$

We shall now show that $P(m + 1)$ is true.

$$\text{i.e., } \frac{1}{1.2.3} + \frac{1}{2.3.4} + \dots + \frac{1}{m(m+1)(m+2)} + \frac{1}{(m+1)(m+2)(m+3)} = \frac{(m+1)(m+4)}{4(m+2)(m+3)}$$

Now,

$$\begin{aligned} & \frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots + \frac{1}{m(m+1)(m+2)} + \frac{1}{(m+1)(m+2)(m+3)} \\ &= \frac{m(m+3)}{4(m+1)(m+2)} + \frac{1}{(m+1)(m+2)(m+3)} \quad [\text{Using (i)}] \end{aligned}$$

$$= \frac{m(m+3)^2 + 4}{4(m+1)(m+2)(m+3)}$$

$$= \frac{m^3 + 6m^2 + 9m + 4}{4(m+1)(m+2)(m+3)} = \frac{(m+1)^2(m+4)}{4(m+1)(m+2)(m+3)} = \frac{(m+1)(m+4)}{4(m+2)(m+3)}$$

$\therefore P(m + 1)$ is true.

Hence, $P(n)$ is true for all $n \in N$.

EXAMPLE 13 If x and y are any two distinct integers, then prove by mathematical induction that $(x^n - y^n)$ is divisible by $(x - y)$ for all $n \in N$.

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : (x^n - y^n) \text{ is divisible by } (x - y)$$

STEP I $P(1) : (x^1 - y^1)$ is divisible by $(x - y)$.

$$\therefore x^1 - y^1 = (x - y) \text{ is divisible by } (x - y).$$

$\therefore P(1)$ is true

STEP II Let $P(m)$ be true. Then,

$(x^m - y^m)$ is divisible by $(x - y)$

$$\Rightarrow (x^m - y^m) = \lambda(x - y), \text{ for some } \lambda \in \mathbb{Z} \quad \dots(i)$$

We shall now show that $P(m+1)$ is true. For this it is sufficient to show that $(x^{m+1} - y^{m+1})$ is divisible by $(x - y)$.

Now,

$$\begin{aligned} x^{m+1} - y^{m+1} &= x^{m+1} - x^m y + x^m y - y^{m+1} \\ &= x^m(x - y) + y(x^m - y^m) \\ &= x^m(x - y) + y\lambda(x - y) \quad [\text{Using (i)}] \\ &= (x - y)(x^m + y\lambda), \text{ which is divisible by } (x - y) \end{aligned}$$

So, $P(m+1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true

Hence, by the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$.

i.e. $(x^n - y^n)$ is divisible by $(x - y)$ for all $n \in \mathbb{N}$.

EXAMPLE 14 Using principle of mathematical induction, prove that $x^{2n} - y^{2n}$ is divisible by $x + y$ for all $n \in \mathbb{N}$.

SOLUTION Let $P(n)$ be the statement given by

$P(n) : (x^{2n} - y^{2n})$ is divisible by $(x + y)$.

STEP I $P(1) : (x^2 - y^2)$ is divisible by $(x + y)$.

$$\therefore (x^2 - y^2) = (x - y)(x + y), \text{ which is divisible by } (x + y)$$

So, $P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$x^{2m} - y^{2m}$ is divisible by $(x + y)$

$$\Rightarrow x^{2m} - y^{2m} = \lambda(x + y) \quad \dots(i)$$

We shall now show that $P(m+1)$ is true i.e., $x^{2m+2} - y^{2m+2}$ is divisible by $(x + y)$.

Now,

$$\begin{aligned} x^{2m+2} - y^{2m+2} &= x^{2m+2} - x^{2m} y^2 + x^{2m} y^2 - y^{2m+2} \\ \Rightarrow x^{2m+2} - y^{2m+2} &= x^{2m}(x^2 - y^2) + y^2(x^{2m} - y^{2m}) \\ \Rightarrow x^{2m+2} - y^{2m+2} &= x^{2m}(x^2 - y^2) + y^2\lambda(x + y) \quad [\text{Using (i)}] \\ \Rightarrow x^{2m+2} - y^{2m+2} &= (x + y) \left\{ x^{2m}(x - y) + \lambda y^2 \right\} \end{aligned}$$

Clearly, it is divisible by $(x + y)$.

$\therefore P(m+1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction $P(n)$ is true for all $n \in \mathbb{N}$ i.e., $x^{2n} - y^{2n}$ is divisible by $(x + y)$ for all $n \in \mathbb{N}$.

EXAMPLE 15 Using principle of mathematical induction, prove that

(i) $41^n - 14^n$ is a multiple of 27

(ii) $7^n - 3^n$ is divisible by 4.

SOLUTION (i) Let $P(n)$ be the statement given by

$P(n) : 41^n - 14^n$ is a multiple of 27.

STEP I $P(1) : 41^1 - 14^1$ is a multiple of 27.

$\therefore 41^1 - 14^1 = 41 - 14 = 27$, which is a multiple of 27.

So, $P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$41^m - 14^m$ is a multiple of 27

$\Rightarrow 41^m - 14^m = 27\lambda$ for some $\lambda \in N$

...(i)

Now, $41^{m+1} - 14^{m+1} = 41^{m+1} - 41 \times 14^m + 41 \times 14^m - 14^{m+1}$

$\Rightarrow 41^{m+1} - 14^{m+1} = 41(41^m - 14^m) + (41 - 14)14^m$

$\Rightarrow 41^{m+1} - 14^{m+1} = 41 \times 27\lambda + 27 \times 14^m$

[Using (i)]

$\Rightarrow 41^{m+1} - 14^{m+1} = 27(41\lambda + 14^m)$, which is a multiple of 27.

$\therefore P(m+1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, $P(n)$ is true for all $n \in N$.

(ii) Proceed as in (i).

EXAMPLE 16 Using the principle of mathematical induction, prove that $(2^{3n} - 1)$ is divisible by 7 for all $n \in N$. [NCERT EXEMPLAR]

SOLUTION Let $P(n)$ be the statement given by

$P(n) : 2^{3n} - 1$ is divisible by 7

STEP I $P(1) : 2^{3 \times 1} - 1$ is divisible by 7.

Clearly, $2^{3 \times 1} - 1 = 8 - 1 = 7$, which is divisible by 7.

So, $P(1)$ is true

STEP II Let $P(m)$ be true. Then,

$2^{3m} - 1$ is divisible by 7.

$\Rightarrow 2^{3m} - 1 = 7\lambda$, for some $\lambda \in N$

...(i)

We shall now show that $P(m+1)$ is true. For this we have to show that $2^{3(m+1)} - 1$ is divisible by 7.

Now,

$$2^{3(m+1)} - 1 = 2^{3m} \times 2^3 - 1 = (7\lambda + 1)2^3 - 1$$

[Using (i)]

$$= 56\lambda + 8 - 1 = 7(8\lambda + 1), \text{ which is divisible by 7}$$

$\therefore P(m+1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true

Hence, by the principle of mathematical induction, $P(n)$ is true for all $n \in N$ i.e. $2^{3n} - 1$ is divisible by 7 for all $n \in N$.

EXAMPLE 17 Prove by the principle of induction that for all $n \in N$, $(10^{2n-1} + 1)$ is divisible by 11.

SOLUTION Let $P(n)$ be the statement given by

$P(n) : 10^{2n-1} + 1$ is divisible by 11

STEP I We have

$P(1) : 10^{2 \times 1 - 1} + 1$ is divisible by 11.

Since $10^{2 \times 1 - 1} + 1 = 11$, which is divisible by 11.

So, $P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$10^{2m-1} + 1$ is divisible by 11

$\Rightarrow 10^{2m-1} + 1 = 11\lambda$, for some $\lambda \in N$

...(i)

We shall now show that $P(m+1)$ is true. For this we have to show that $10^{2(m+1)} + 1$ is divisible by 11.

Now, $10^{2(m+1)-1} + 1 = 10^{2m+1} + 1 = 10^{2m-1} \times 10^2 + 1$

$\Rightarrow 10^{2(m+1)-1} + 1 = (11\lambda - 1)100 + 1$

[Using (i)]

$\Rightarrow 10^{2(m+1)-1} + 1 = 1100\lambda - 99 = 11(100\lambda - 9) = 11\mu$, where $\mu = 100\lambda - 9 \in N$

$\Rightarrow 10^{2(m+1)} - 1$ is divisible by 11

$\Rightarrow P(m+1)$ is true

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true

Hence, by the principle of mathematical induction $P(m)$ is true for all $n \in N$ i.e. $10^{2n-1} + 1$ is divisible by 11 for all $n \in N$.

EXAMPLE 18 Prove that for $n \in N$, $10^n + 3 \cdot 4^{n+2} + 5$ is divisible by 9.

SOLUTION Let $P(n)$ be the statement given by

$P(n) : 10^n + 3 \cdot 4^{n+2} + 5$ is divisible by 9

STEP I $P(1) : 10^1 + 3(4^{1+2}) + 5$ is divisible by 9.

$\because 10^1 + 3(4^{1+2}) + 5 = 10 + 192 + 5 = 207$, which is divisible by 9

$\therefore P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$10^m + 3(4^{m+2}) + 5$ is divisible by 9

$\Rightarrow 10^m + 3(4^{m+2}) + 5 = 9\lambda$, $\lambda \in N$

...(i)

We shall now show that $P(m+1)$ is true for which we have to show that $10^{(m+1)} + 3(4^{m+3}) + 5$ is divisible by 9.

Now,

$$10^{m+1} + 3(4^{m+3}) + 5 = 10^m(10) + 3(4^{m+3}) + 5$$

$$= \{9\lambda - 3(4^{m+2}) - 5\} \times 10 + 3 \times 4^{m+3} + 5$$

[Using (i)]

$$= 90\lambda - 30 \times 4^{m+2} - 50 + 3 \times 4 \times 4^{m+2} + 5$$

$$= 90\lambda - 30 \times 4^{m+2} + 12 \times 4^{m+2} - 45$$

$$= 90\lambda - 18 \times 4^{m+2} - 45$$

$$= 9(10\lambda - 2 \times 4^{m+2} - 5) = 9\mu$$
, where $\mu = 10\lambda - 2 \times 4^{m+2} - 5$

$\Rightarrow 10^{m+1} + 3 \cdot 4^{m+3} + 5$ is divisible by 9

$\Rightarrow P(m+1)$ is true

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction $P(n)$ is true for all $n \in N$.

EXAMPLE 19 Prove by induction that the sum of the cubes of three consecutive natural numbers is divisible by 9.

SOLUTION Let $P(n)$ be the statement given by

$P(n)$: Sum of the cubes of three consecutive natural numbers starting from n is divisible by 9.

STEP I $P(1)$: Sum of the cubes of first three consecutive natural numbers is divisible by 9.

Since $1^3 + 2^3 + 3^3 = 36$, which is divisible by 9.

$\therefore P(1)$ is true.

STEP II Let $P(m)$ be true. Then, sum of the cubes of three consecutive natural numbers starting with m is divisible by 9.

i.e. $m^3 + (m+1)^3 + (m+2)^3$ is divisible by 9

$\Rightarrow m^3 + (m+1)^3 + (m+2)^3 = 9\lambda, \lambda \in N$

We shall now show that $P(m+1)$ is true for which we have to show that

$(m+1)^3 + (m+2)^3 + (m+3)^3$ is divisible by 9.

Now, $(m+1)^3 + (m+2)^3 + (m+3)^3$

$$= (m+1)^3 + (m+2)^3 + m^3 + 9m^2 + 27m + 27$$

$$= m^3 + (m+1)^3 + (m+2)^3 + 9(m^2 + 3m + 3)$$

$$= 9\lambda + 9(m^2 + 3m + 3)$$

[Using (i)]

$$= 9(\lambda + m^2 + 3m + 3), \text{ which is divisible by 9.}$$

$\therefore P(m+1)$ is true

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction $P(n)$ is true for all $n \in N$.

EXAMPLE 20 Using principle of mathematical induction prove that $4^n + 15n - 1$ is divisible by 9 for all natural numbers n .

SOLUTION Let $P(n)$ be the statement given by

$P(n)$: $4^n + 15n - 1$ is divisible by 9

STEP I $P(1)$: $4^1 + 15 \times 1 - 1$ is divisible by 9.

$\therefore 4^1 + 15 \times 1 - 1 = 18$, which is divisible by 9

$\therefore P(1)$ is true

STEP II Let $P(m)$ be true. Then,

$4^m + 15m - 1$ is divisible by 9

$\Rightarrow 4^m + 15m - 1 = 9\lambda$, for some $\lambda \in N$

We shall now show that $P(m+1)$ is true, for this we have to show that $4^{m+1} + 15(m+1) - 1$ is divisible by 9.

Now,

$$4^{m+1} + 15(m+1) - 1 = 4^m \cdot 4 + 15(m+1) - 1$$

$$= (9\lambda - 15m + 1) \times 4 + 15(m+1) - 1$$

$$= 36\lambda - 45m + 18 = 9(4\lambda - 5m + 2), \text{ which is divisible by 9.}$$

$\therefore P(m+1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true

Hence, by the principle of mathematical induction $P(n)$ is true for all $n \in N$ i.e., $4^n + 15n - 1$ is divisible by 9.

EXAMPLE 21 Prove that: $2 \cdot 7^n + 3 \cdot 5^n - 5$ is divisible by 24, for all $n \in N$.

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : 2 \cdot 7^n + 3 \cdot 5^n - 5 \text{ is divisible by } 24.$$

STEP I We have,

$$P(1) : 2 \times 7^1 + 3 \times 5^1 - 5 \text{ is divisible by } 24$$

$$\therefore 2 \times 7^1 + 3 \times 5^1 - 5 = 14 + 15 - 5 = 24, \text{ which is divisible by } 24.$$

$$\therefore P(1) \text{ is true.}$$

STEP II Let $P(m)$ be true. Then,

$$2 \times 7^m + 3 \times 5^m - 5 \text{ is divisible by } 24$$

$$\Rightarrow 2 \times 7^m + 3 \times 5^m - 5 = 24\lambda \text{ for some } \lambda \in N$$

$$\Rightarrow 3 \times 5^m = 24\lambda + 5 - 2 \times 7^m \quad \dots(i)$$

$$\text{Now, } 2 \times 7^{m+1} + 3 \times 5^{m+1} - 5$$

$$= 2 \times 7^{m+1} + (3 \times 5^m) 5 - 5$$

$$= 2 \times 7^{m+1} + (24\lambda + 5 - 2 \times 7^m) 5 - 5 \quad [\text{Using (i)}]$$

$$= 2 \times 7^{m+1} + 120\lambda + 25 - 10 \times 7^m - 5$$

$$= (2 \times 7^{m+1} - 10 \times 7^m) + 120\lambda + 20$$

$$= (2 \times 7 \times 7^m - 10 \times 7^m) + 120\lambda + 24 - 4$$

$$= (14 - 10) 7^m - 4 + 24(5\lambda + 1)$$

$$= 4(7^m - 1) + 24(5\lambda + 1)$$

$$= 4 \times 6\mu + 24(5\lambda + 1) \quad [\because 7^m - 1 \text{ is a multiple of } 6 \text{ for all } m \in N \therefore 7^m - 1 = 6\mu, \mu \in N]$$

$$= 24(\mu + 5\lambda + 1), \text{ which is divisible by } 24.$$

$$\therefore P(m+1) \text{ is true.}$$

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction, $P(n)$ is true for all $n \in N$.

EXAMPLE 22 Prove that :

$$(i) \left(1 + \frac{1}{1}\right) \left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{3}\right) \dots \left(1 + \frac{1}{n}\right) = (n+1) \text{ for all } n \in N.$$

$$(ii) \left(1 + \frac{1}{3}\right) \left(1 + \frac{5}{4}\right) \left(1 + \frac{7}{9}\right) \dots \left(1 + \frac{2n+1}{n^2}\right) = (n+1)^2 \text{ for all } n \in N.$$

SOLUTION (i) Let $P(n)$ be the statement given by

$$P(n) : \left(1 + \frac{1}{1}\right) \left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{3}\right) \dots \left(1 + \frac{1}{n}\right) = n+1$$

STEP I We have,

$$P(1) : \left(1 + \frac{1}{1}\right) = (1+1)$$

$$\therefore \left(1 + \frac{1}{1}\right) = 2 = (1+1)$$

$$\therefore P(1) \text{ is true.}$$

STEP II Let $P(m)$ be true. Then,

$$\left(1 + \frac{1}{1}\right)\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right)\dots\left(1 + \frac{1}{m}\right) = m + 1 \quad \dots(i)$$

Now,

$P(m)$ is true

$$\Rightarrow \left(1 + \frac{1}{1}\right)\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right)\dots\left(1 + \frac{1}{m}\right) = (m + 1) \quad [\text{From (i)}]$$

$$\begin{aligned} \Rightarrow & \left(1 + \frac{1}{1}\right)\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right)\dots\left(1 + \frac{1}{m}\right) \left(1 + \frac{1}{m+1}\right) \\ &= (m + 1) \left(1 + \frac{1}{m+1}\right) \quad \left[\text{Multiplying both sides by } \left(1 + \frac{1}{m+1}\right) \right] \\ &= \frac{(m + 1)(m + 2)}{m + 1} = m + 2 \end{aligned}$$

$\therefore P(m + 1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m + 1)$ is true.

Hence, by the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$.

(ii) Let $P(n)$ be the statement given by

$$P(n) : \left(1 + \frac{3}{1}\right)\left(1 + \frac{5}{4}\right)\left(1 + \frac{7}{9}\right)\dots\left(1 + \frac{2n+1}{n^2}\right) = (n + 1)^2$$

STEP I We have,

$$P(1) : \left(1 + \frac{3}{1}\right) = (1 + 1)^2$$

$$\therefore 1 + \frac{3}{1} = 1 + 3 = 4 = (1 + 1)^2$$

$\therefore P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$\left(1 + \frac{3}{1}\right)\left(1 + \frac{5}{4}\right)\left(1 + \frac{7}{9}\right)\dots\left(1 + \frac{2m+1}{m^2}\right) = (m + 1)^2 \quad \dots(i)$$

We shall now prove that $P(m + 1)$ is true.

$$\text{i.e.} \quad \left(1 + \frac{3}{1}\right)\left(1 + \frac{5}{4}\right)\left(1 + \frac{7}{9}\right)\dots\left(1 + \frac{2m+1}{m^2}\right) \left(1 + \frac{2(m+1)+1}{(m+1)^2}\right) = \{(m+1) + 1\}^2$$

Now,

$P(m)$ is true

$$\Rightarrow \left(1 + \frac{3}{1}\right)\left(1 + \frac{5}{4}\right)\left(1 + \frac{7}{9}\right)\dots\left(1 + \frac{2m+1}{m^2}\right) = (m + 1)^2 \quad [\text{From (i)}]$$

$$\begin{aligned} \Rightarrow & \left(1 + \frac{3}{1}\right)\left(1 + \frac{5}{4}\right)\left(1 + \frac{7}{9}\right)\dots\left(1 + \frac{2m+1}{m^2}\right) \left(1 + \frac{2m+3}{(m+1)^2}\right) \\ &= (m + 1)^2 \left(1 + \frac{2m+3}{(m+1)^2}\right) \quad \left[\text{Multiplying both sides by } 1 + \frac{2m+3}{(m+1)^2} \right] \\ &= (m + 1)^2 \left\{ \frac{(m+1)^2 + 2m+3}{(m+1)^2} \right\} = (m^2 + 4m + 4) = (m + 2)^2 = \{(m+1) + 1\}^2 \end{aligned}$$

$\therefore P(m + 1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction, $P(n)$ is true for all $n \in N$.

EXAMPLE 23 Prove by induction that $4 + 8 + 12 + \dots + 4n = 2n(n+1)$ for all $n \in N$.

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : 4 + 8 + 12 + \dots + 4n = 2n(n+1)$$

STEP I $P(1) : 4 = 2 \times 1 \times (1+1)$, which is true.

$\therefore P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$4 + 8 + 12 + \dots + 4m = 2m(m+1) \quad \dots(i)$$

We shall now show that $P(m+1)$ is true

$$\text{i.e. } 4 + 8 + \dots + 4m + 4(m+1) = 2(m+1)\{(m+1)+1\}.$$

Now,

$$4 + 8 + \dots + 4m + 4(m+1)$$

$$= 2m(m+1) + 4(m+1)$$

[Using (i)]

$$= (m+1)(2m+4) = 2(m+1)(m+2) = 2(m+1)\{(m+1)+1\}$$

$\therefore P(m+1)$ is true.

Thus $P(m)$ is true $\Rightarrow P(m+1)$ is true

Hence, by induction $P(n)$ is true for all $n \in N$.

LEVEL-2

EXAMPLE 24 For all positive integer n , prove that $\frac{n^7}{7} + \frac{n^5}{5} + \frac{2}{3}n^3 - \frac{n}{105}$ is an integer

SOLUTION Let $P(n)$ be the statement given by

$$P(1) : \frac{n^7}{7} + \frac{n^5}{5} + 2\frac{n^3}{3} - \frac{n}{105} \text{ is an integer}$$

STEP I $P(1) : \frac{1}{7} + \frac{1}{5} + \frac{2}{3} - \frac{1}{105}$ is an integer.

$$\text{Since } \frac{1}{7} + \frac{1}{5} + \frac{2}{3} - \frac{1}{105} = \frac{15 + 21 + 70 - 1}{105} = 1, \text{ which is an integer.}$$

So, $P(1)$ is true.

STEP II Let $P(m)$ be true. Then, $\frac{m^7}{7} + \frac{m^5}{5} + \frac{2m^3}{3} - \frac{m}{105}$ is an integer

$$\text{Let } \frac{m^7}{7} + \frac{m^5}{5} + \frac{2m^3}{3} - \frac{m}{105} = \lambda, \lambda \in Z$$

$\dots(i)$

We shall now show that $P(m+1)$ is true for which we have to show that

$$\frac{(m+1)^7}{7} + \frac{(m+1)^5}{5} + \frac{2(m+1)^3}{3} - \frac{(m+1)}{105} \text{ is an integer.}$$

$$\text{Now, } \frac{(m+1)^7}{7} + \frac{(m+1)^5}{5} + \frac{2(m+1)^3}{3} - \frac{(m+1)}{105}$$

$$= \frac{1}{7}(m^7 + 7m^6 + 21m^5 + 35m^4 + 35m^3 + 21m^2 + 7m + 1)$$

$$+ \frac{1}{5}(m^5 + 5m^4 + 10m^3 + 10m^2 + 5m + 1) + \frac{2}{3}(m^3 + 3m^2 + 3m + 1) - \frac{m}{105} - \frac{1}{105}$$

$$= \left\{ \frac{m^7}{7} + \frac{m^5}{5} + 2 \frac{m^3}{3} - \frac{m}{105} \right\} + m^6 + 3m^5 + 6m^4 + 7m^3 + 7m^2 + 4m + 1$$

$$= \lambda + m^6 + 3m^5 + 6m^4 + 7m^3 + 7m^2 + 4m + 1$$

[Using (i)]

$$= \text{an integer}$$

$\therefore P(m+1)$ is true

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction $P(n)$ is true for all $n \in N$.

i.e. $\frac{n^7}{7} + \frac{n^5}{5} + 2 \frac{n^3}{3} - \frac{n}{105}$ is an integer.

EXAMPLE 25 Prove by the principle of mathematical induction that $\frac{n^5}{5} + \frac{n^3}{3} + \frac{7n}{15}$ is a natural number for all $n \in N$. [NCERT EXEMPLAR]

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : \frac{n^5}{5} + \frac{n^3}{3} + \frac{7n}{15} \text{ is a natural number}$$

STEP I $P(1) : \frac{1}{5} + \frac{1}{3} + \frac{7}{15}$ is a natural number.

$$\therefore \frac{1}{5} + \frac{1}{3} + \frac{7}{15} = \frac{15}{15} = 1, \text{ which is a natural number.}$$

So, $P(1)$ is true.

STEP II Let $P(m)$ be true.

$$\text{Then, } \frac{m^5}{5} + \frac{m^3}{3} + \frac{7m}{15} \text{ is a natural number. Let } \frac{m^5}{5} + \frac{m^3}{3} + \frac{7m}{15} = \lambda \quad \dots(i)$$

We shall now show that $P(m+1)$ is true, for which it is sufficient to prove that

$$\frac{(m+1)^5}{5} + \frac{(m+1)^3}{3} + \frac{7(m+1)}{15} \text{ is a natural number.}$$

$$\text{Now, } \frac{(m+1)^5}{5} + \frac{(m+1)^3}{3} + \frac{7(m+1)}{15}$$

$$= \frac{1}{5} (m^5 + 5m^4 + 10m^3 + 10m^2 + 5m + 1) + \frac{1}{3} (m^3 + 3m^2 + 3m + 1) + \frac{7}{15} m + \frac{7}{15}$$

$$= \left(\frac{m^5}{5} + \frac{m^3}{3} + \frac{7}{15} m \right) + (m^4 + 2m^3 + 3m^2 + 2m) + \frac{1}{5} + \frac{1}{3} + \frac{7}{15}$$

$$= \lambda + m^4 + 2m^3 + 3m^2 + 2m + 1$$

[Using (i)]

$$= \text{an integer}$$

$\therefore P(m+1)$ is true

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction $P(n)$ is true for all $n \in N$.

i.e. $\frac{n^5}{5} + \frac{n^3}{3} + \frac{7}{15} n$ is a natural number for all $n \in N$.

EXAMPLE 26 Prove by the principle of mathematical induction that for all $n \in N$, 3^{2n} when divided by 8, the remainder is always 1.

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : 3^{2n} \text{ when divided by 8, the remainder is 1}$$

or, $P(n) : 3^{2n} = 8\lambda + 1 \text{ for some } \lambda \in N$

STEP I $P(1) : 3^2 = 8\lambda + 1 \text{ for some } \lambda \in N.$

$$\therefore 3^2 = 8 \times 1 + 1 = 8\lambda + 1, \text{ where } \lambda = 1$$

$$\therefore P(1) \text{ is true}$$

STEP II Let $P(m)$ be true. Then,

$$3^{2m} = 8\lambda + 1 \text{ for some } \lambda \in N$$

...(i)

We shall now show that $P(m+1)$ is true for which we have to show that $3^{2(m+1)}$ when divided by 8, the remainder is 1 i.e. $3^{2(m+1)} = 8\mu + 1$ for some $\mu \in N$.

$$\begin{aligned} \text{Now, } 3^{2(m+1)} &= 3^{2m} \times 3^2 = (8\lambda + 1) \times 9 && [\text{Using (i)}] \\ &= 72\lambda + 9 = 72\lambda + 8 + 1 = 8(9\lambda + 1) + 1 = 8\mu + 1, \text{ where } \mu = 9\lambda + 1 \in N \end{aligned}$$

$$\Rightarrow P(m+1) \text{ is true}$$

$$\text{Thus, } P(m) \text{ is true} \Rightarrow P(m+1) \text{ is true.}$$

Hence, by the principle of mathematical induction $P(n)$ is true for all $n \in N$ i.e. 3^{2n} when divided by 8 the remainder is always 1.

EXAMPLE 27 Prove by the principle of mathematical induction that $n < 2^n$ for all $n \in N$.

SOLUTION Let $P(n)$ be the statement given by $P(n) : n < 2^n$.

STEP I $P(1) : 1 < 2^1$

$$\therefore 1 < 2^1$$

$$\therefore P(1) \text{ is true}$$

STEP II Let $P(m)$ be true. Then, $m < 2^m$

We shall now show that $P(m+1)$ is true for which we will have to prove that $(m+1) < 2^{m+1}$.

Now,

$$P(m) \text{ is true}$$

$$\Rightarrow m < 2^m$$

$$\Rightarrow 2m < 2 \cdot 2^m$$

$$\Rightarrow 2m < 2^{m+1}$$

$$\Rightarrow (m+m) < 2^{m+1}$$

$$\Rightarrow m+1 \leq m+m < 2^{m+1}$$

$$[\because 1 \leq m \therefore m+1 \leq m+m]$$

$$\Rightarrow (m+1) < 2^{m+1}$$

$$\Rightarrow P(m+1) \text{ is true}$$

$$\text{Thus, } P(m) \text{ is true} \Rightarrow P(m+1) \text{ is true.}$$

So, by the principle of mathematical induction $P(n)$ is true for all $n \in N$ i.e. $n < 2^n$ for all $n \in N$.

EXAMPLE 28 Prove by induction the inequality $(1+x)^n \geq 1+nx$ whenever x is positive and n is a positive integer.

SOLUTION Let $P(n)$ be the statement given by $P(n) : (1+x)^n \geq 1+nx$

STEP I $P(1) : (1+x)^1 \geq 1+1(x)$

$$\therefore (1+x)^1 \geq 1 + 1(x)$$

$$\therefore P(1) \text{ is true}$$

STEP II Let $P(m)$ be true. Then,

$$(1+x)^m \geq 1 + mx \quad \dots(i)$$

We shall now prove that $P(m+1)$ is true whenever $P(m)$ is true. For this we have to show that $(1+x)^{m+1} \geq 1 + (m+1)x$.

Now, $P(m)$ is true

$$\Rightarrow (1+x)^m \geq 1 + mx$$

$$\Rightarrow (1+x)(1+x)^m \geq (1+x)(1+mx) \quad [\text{Multiplying both sides by } (1+x)]$$

$$\Rightarrow (1+x)^{m+1} \geq 1 + (m+1)x + mx^2$$

$$\Rightarrow (1+x)^{m+1} \geq 1 + (m+1)x + mx^2 \geq 1 + (m+1)x \quad [\because mx^2 \geq 0]$$

$$\Rightarrow (1+x)^{m+1} \geq 1 + (m+1)x$$

$$\Rightarrow P(m+1) \text{ is true}$$

Hence, by the principle of induction, $P(n)$ is true for all $n \in N$ i.e. $(1+x)^n \geq 1 + nx$ for all $n \in N$.

EXAMPLE 29 Prove by induction that $(2n+7) < (n+3)^2$ for all natural numbers n . Using this, prove by induction that $(n+3)^2 \leq 2^{n+3}$ for all $n \in N$.

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : (2n+7) < (n+3)^2$$

$$\text{STEP I} \quad P(1) : (2 \times 1 + 7) < (1+3)^2$$

$$\therefore (2 \times 1 + 7) = 9 < (1+3)^2$$

$$\therefore P(1) \text{ is true}$$

$$\text{STEP II} \quad \text{Let } P(m) \text{ be true. Then, } 2m+7 < (m+3)^2. \quad \dots(i)$$

We shall now show that $P(m+1)$ is true whenever $P(m)$ is true. For this we have to show that $2(m+1)+7 < (m+1+3)^2$.

Now,

$$P(m) \text{ is true}$$

$$\Rightarrow 2m+7 < (m+3)^2$$

$$\Rightarrow 2m+7+2 < (m+3)^2+2$$

$$\Rightarrow 2(m+1)+7 < m^2+6m+11$$

$$\Rightarrow 2(m+1)+7 < m^2+6m+11 < m^2+8m+16$$

$$\Rightarrow 2(m+1)+7 < (m+4)^2$$

$$\Rightarrow \{2(m+1)+7\} < \{(m+1)+3\}^2$$

$$\Rightarrow P(m+1) \text{ is true}$$

Hence, by the principle of mathematical induction, $P(n)$ is true for all $n \in N$.

Now, let $P'(n)$ be the statement given by $P'(n) : (n+3)^2 \leq 2^{n+3}$

$$\text{STEP I} \quad P'(1) : (1+3)^2 \leq 2^{1+3}$$

$$\therefore (1+3)^2 = 16 \leq 2^{1+3}$$

$$\therefore P'(1) \text{ is true}$$

STEP II Let $P'(m)$ be true. Then, $(m+3)^2 \leq 2^{m+3}$.

We shall now show that $P'(m+1)$ is true whenever $P'(m)$ is true. For this we have to show that $\{(m+1)+3\}^2 \leq 2^{(m+1)+3}$.

Now, $P'(m)$ is true

$$\Rightarrow (m+3)^2 \leq 2^{m+3}$$

$$\Rightarrow (m+3)^2 + (2m+7) \leq 2^{m+3} + (2m+7)$$

$$\Rightarrow (m+4)^2 \leq 2^{m+3} + (m+3)^2 \quad [\because 2m+7 < (m+3)^2 \therefore 2^{m+3} + (2m+7) < 2^{m+3} + (m+3)^2]$$

$$\Rightarrow (m+4)^2 \leq 2^{m+3} + 2^{m+3} \quad [\because (m+3)^2 \leq 2^{m+3} \Rightarrow (m+3)^2 + 2^{m+3} \leq 2^{m+3} + 2^{m+3}]$$

$$\Rightarrow (m+4)^2 \leq 2 \cdot 2^{m+3}$$

$$\Rightarrow (m+4)^2 \leq 2^{m+4}$$

$$\Rightarrow \{(m+1)+3\}^2 \leq 2^{(m+1)+3}$$

$$\Rightarrow P'(m+1) \text{ is true}$$

Hence, by the principle of mathematical induction, $P'(n)$ is true for all $n \in N$ i.e. $(n+3)^2 \leq 2^{n+3}$ for all $n \in N$.

EXAMPLE 30 Prove that: $1^2 + 2^2 + 3^2 + \dots + n^2 > \frac{n^3}{3}$ for all $n \in N$.

SOLUTION Let $P(n)$ be the statement given by

$$P(n): 1^2 + 2^2 + 3^2 + \dots + n^2 > \frac{n^3}{3}$$

STEP I $P(1): 1^2 > \frac{1^3}{3}$

$$\because 1^2 = 1 > \frac{1}{3} = \frac{1^3}{3}$$

$$\therefore P(1) \text{ is true.}$$

STEP II Let $P(n)$ be true for $n=m$. Then,

$$1^2 + 2^2 + 3^2 + \dots + m^2 > \frac{m^3}{3}$$

...(i)

We shall now prove that $P(m+1)$ is true.

$$\text{i.e. } 1^2 + 2^2 + 3^2 + \dots + m^2 + (m+1)^2 > \frac{(m+1)^3}{3}$$

Now, $P(m)$ is true

$$\Rightarrow 1^2 + 2^2 + 3^2 + \dots + m^2 > \frac{m^3}{3}$$

$$\Rightarrow 1^2 + 2^2 + 3^2 + \dots + m^2 + (m+1)^2 > \frac{m^3}{3} + (m+1)^2$$

$$\Rightarrow 1^2 + 2^2 + 3^2 + \dots + m^2 + (m+1)^2 > \frac{1}{3}(m^3 + 3m^2 + 6m + 3)$$

$$\Rightarrow 1^2 + 2^2 + 3^2 + \dots + m^2 + (m+1)^2 > \frac{1}{3} \left\{ (m^3 + 3m^2 + 3m + 1) + (3m + 2) \right\}$$

$$\Rightarrow 1^2 + 2^2 + 3^2 + \dots + m^2 + (m+1)^2 > \frac{1}{3} \left\{ (m+1)^3 + (3m+2) \right\} > \frac{(m+1)^3}{3}$$

$\Rightarrow P(m+1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction, $P(n)$ is true for all $n \in N$.

EXAMPLE 31 Prove that: $1 + 2 + 3 + \dots + n < \frac{(2n+1)^2}{8}$ for all $n \in N$.

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : 1 + 2 + 3 + \dots + n < \frac{(2n+1)^2}{8}$$

STEP I We have,

$$P(1) : 1 < \frac{(2 \times 1 + 1)^2}{8}$$

$$\therefore 1 < \frac{(2 \times 1 + 1)^2}{8} = \frac{9}{8}$$

$\therefore P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$1 + 2 + 3 + \dots + m < \frac{(2m+1)^2}{8} \quad \dots(i)$$

We shall now show that $P(m+1)$ is true.

$$\text{i.e., } 1 + 2 + 3 + \dots + m + (m+1) < \frac{[2(m+1)+1]^2}{8}$$

Now,

$P(m)$ is true

$$\Rightarrow 1 + 2 + 3 + \dots + m < \frac{(2m+1)^2}{8}$$

$$\Rightarrow 1 + 2 + 3 + \dots + m + (m+1) < \frac{(2m+1)^2}{8} + (m+1)$$

$$\Rightarrow 1 + 2 + 3 + \dots + m + (m+1) < \frac{(2m+1)^2 + 8(m+1)}{8}$$

$$\Rightarrow 1 + 2 + 3 + \dots + m + (m+1) < \frac{(4m^2 + 12m + 9)}{8}$$

$$\Rightarrow 1 + 2 + 3 + \dots + m + (m+1) < \frac{(2m+3)^2}{8} = \frac{[2(m+1)+1]^2}{8}$$

$\therefore P(m+1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by the principle of mathematical induction $P(n)$ is true for all $n \in N$.

EXAMPLE 32 Prove by the principle of mathematical induction that for all $n \in N$,

$$\sin \theta + \sin 2\theta + \sin 3\theta + \dots + \sin n\theta = \frac{\sin \left(\frac{n+1}{2} \right) \theta \sin \frac{n\theta}{2}}{\sin \frac{\theta}{2}} \quad \text{[NCERT EXEMPLAR]}$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : \sin \theta + \sin 2\theta + \sin 3\theta + \dots + \sin n\theta = \frac{\sin \left(\frac{n+1}{2} \right) \theta \sin \frac{n\theta}{2}}{\sin \frac{\theta}{2}}$$

STEP I We have, $P(1) : \sin \theta = \frac{\sin \left(\frac{1+1}{2} \right) \theta \sin \left(\frac{1 \times \theta}{2} \right)}{\sin \frac{\theta}{2}}$

$$\therefore \sin \theta = \frac{\sin \left(\frac{1+1}{2} \right) \theta \cdot \sin \left(\frac{1 \times \theta}{2} \right)}{\sin \frac{\theta}{2}}$$

$\therefore P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$\sin \theta + \sin 2\theta + \dots + \sin m\theta = \frac{\sin \left(\frac{m+1}{2} \right) \theta \sin \frac{m\theta}{2}}{\sin \frac{\theta}{2}} \quad \dots(i)$$

We shall now show that $P(m+1)$ is true.

$$\text{i.e.} \quad \sin \theta + \sin 2\theta + \dots + \sin m\theta + \sin (m+1)\theta = \frac{\sin \left\{ \frac{(m+1)+1}{2} \right\} \theta \sin \left(\frac{m+1}{2} \right) \theta}{\sin \frac{\theta}{2}}$$

Now,

$$\begin{aligned} & \sin \theta + \sin 2\theta + \dots + \sin m\theta + \sin (m+1)\theta \\ &= \frac{\sin \left(\frac{m+1}{2} \right) \theta \sin \frac{m\theta}{2}}{\sin \frac{\theta}{2}} + \sin (m+1)\theta \quad \text{[Using (i)]} \\ &= \frac{\sin \left(\frac{m+1}{2} \right) \theta \sin \frac{m\theta}{2}}{\sin \frac{\theta}{2}} + 2 \sin \left(\frac{m+1}{2} \right) \theta \cos \left(\frac{m+1}{2} \right) \theta \\ &= \sin \left(\frac{m+1}{2} \right) \theta \left\{ \frac{\sin \left(\frac{m\theta}{2} \right)}{\sin \frac{\theta}{2}} + 2 \cos \left(\frac{m+1}{2} \right) \theta \right\} \\ &= \sin \left(\frac{m+1}{2} \right) \theta \left\{ \frac{\sin \left(\frac{m\theta}{2} \right) + 2 \sin \frac{\theta}{2} \cos \left(\frac{m+1}{2} \right) \theta}{\sin \frac{\theta}{2}} \right\} \\ &= \sin \left(\frac{m+1}{2} \right) \theta \left\{ \frac{\sin \left(\frac{m\theta}{2} \right) + \sin \left(\frac{m+2}{2} \right) \theta - \sin \frac{m\theta}{2}}{\sin \frac{\theta}{2}} \right\} \end{aligned}$$

$$= \frac{\sin\left(\frac{m+1}{2}\theta\right) \sin\left(\frac{m+2}{2}\theta\right)}{\sin\frac{\theta}{2}} = \frac{\sin\left\{\frac{(m+1)+1}{2}\right\}\theta \sin\left(\frac{m+1}{2}\theta\right)}{\sin\frac{\theta}{2}}$$

$\therefore P(m+1)$ is true

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true.

Hence, by principle of mathematical induction $P(n)$ is true for all $n \in N$.

EXAMPLE 33 Using principle of mathematical induction, prove that

$$\cos \alpha \cos 2\alpha \cos 4\alpha \dots \cos (2^{n-1}\alpha) = \frac{\sin 2^n \alpha}{2^n \sin \alpha} \text{ for all } n \in N. \quad [\text{NCERT EXEMPLAR}]$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n) : \cos \alpha \cos 2\alpha \cos 4\alpha \dots \cos (2^{n-1}\alpha) = \frac{\sin (2^n \alpha)}{2^n \sin \alpha}$$

STEP I $P(1) : \cos \alpha = \frac{\sin (2^1 \alpha)}{2^1 \sin \alpha}$

$$\therefore \frac{\sin (2^1 \alpha)}{2^1 \sin \alpha} = \frac{\sin 2\alpha}{2 \sin \alpha} = \frac{2 \sin \alpha \cos \alpha}{2 \sin \alpha} = \cos \alpha$$

$\therefore P(1)$ is true.

STEP II Let $P(m)$ be true. Then,

$$\cos \alpha \cos 2\alpha \cos 4\alpha \dots \cos (2^{m-1}\alpha) = \frac{\sin (2^m \alpha)}{2^m \sin \alpha} \quad \dots(i)$$

We shall now show that $P(m+1)$ is true. For this we have to show that

$$\cos \alpha \cos 2\alpha \cos 2^2 \alpha \dots \cos (2^{m-1}\alpha) \cos (2^m \alpha) = \frac{\sin (2^{m+1} \alpha)}{2^{m+1} \sin \alpha}$$

We have,

$$\begin{aligned} & \cos \alpha \cos 2\alpha \cos 2^2 \alpha \dots \cos (2^{m-1}\alpha) \cos (2^m \alpha) \\ &= \{\cos \alpha \cos 2\alpha \cos 2^2 \alpha \dots \cos (2^{m-1}\alpha)\} \cos (2^m \alpha) \\ &= \frac{\sin (2^m \alpha)}{2^m \sin \alpha} \times \cos (2^m \alpha) \quad [\text{Using (i)}] \\ &= \frac{2 \sin (2^m \alpha) \cos (2^m \alpha)}{2^{m+1} \sin \alpha} = \frac{\sin (2 \cdot 2^m \alpha)}{2^{m+1} \sin \alpha} = \frac{\sin (2^{m+1} \alpha)}{2^{m+1} \sin \alpha} \end{aligned}$$

$\therefore P(m+1)$ is true.

Thus, $P(m)$ is true $\Rightarrow P(m+1)$ is true

Hence, by the principle of mathematical induction $P(n)$ is true for all $n \in N$.

Type II PROBLEMS BASED UPON SECOND PRINCIPLE OF MATHEMATICAL INDUCTION

EXAMPLE 34 Let $U_1 = 1, U_2 = 1$ and $U_{n+2} = U_{n+1} + U_n$ for $n \geq 1$. Use mathematical induction to show that:

$$U_n = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right\} \text{ for all } n \geq 1.$$

SOLUTION Let $P(n)$ be the statement given by

$$P(n): U_n = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right\}$$

We have,

$$U_1 = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^1 - \left(\frac{1-\sqrt{5}}{2} \right)^1 \right\} = 1$$

and,

$$U_2 = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^2 - \left(\frac{1-\sqrt{5}}{2} \right)^2 \right\} = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+5+2\sqrt{5}}{4} \right) - \left(\frac{1+5-2\sqrt{5}}{4} \right) \right\} = 1$$

$\therefore P(1)$ and $P(2)$ are true.

Let $P(n)$ be true for all $n \leq m$.

$$\text{i.e. } U_n = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right\} \text{ for all } n \leq m \quad \dots(i)$$

We shall now show that $P(n)$ is true for $n = m+1$.

$$\text{i.e. } U_{m+1} = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^{m+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{m+1} \right\}$$

We have,

$$U_{n+2} = U_{n+1} + U_n \text{ for } n \geq 1$$

$$\Rightarrow U_{m+1} = U_m + U_{m-1} \text{ for } m \geq 2 \quad [\text{On replacing } n \text{ by } (m-1)]$$

$$\Rightarrow U_{m+1} = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^m - \left(\frac{1-\sqrt{5}}{2} \right)^m \right\} + \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^{m-1} - \left(\frac{1-\sqrt{5}}{2} \right)^{m-1} \right\}$$

[Using (i)]

$$\Rightarrow U_{m+1} = \frac{1}{\sqrt{5}} \left[\left\{ \left(\frac{1+\sqrt{5}}{2} \right)^m + \left(\frac{1+\sqrt{5}}{2} \right)^{m-1} \right\} - \left\{ \left(\frac{1-\sqrt{5}}{2} \right)^m + \left(\frac{1-\sqrt{5}}{2} \right)^{m-1} \right\} \right]$$

$$\Rightarrow U_{m+1} = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^{m-1} \left(\frac{1+\sqrt{5}}{2} + 1 \right) - \left(\frac{1-\sqrt{5}}{2} \right)^{m-1} \left(\frac{1-\sqrt{5}}{2} + 1 \right) \right\}$$

$$\Rightarrow U_{m+1} = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^{m-1} \left(\frac{3+\sqrt{5}}{2} \right) - \left(\frac{1-\sqrt{5}}{2} \right)^{m-1} \left(\frac{3-\sqrt{5}}{2} \right) \right\}$$

$$\Rightarrow U_{m+1} = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^{m-1} \left(\frac{6+2\sqrt{5}}{4} \right) - \left(\frac{1-\sqrt{5}}{2} \right)^{m-1} \left(\frac{6-2\sqrt{5}}{4} \right) \right\}$$

$$\Rightarrow U_{m+1} = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^{m-1} \left(\frac{1+\sqrt{5}}{2} \right)^2 - \left(\frac{1-\sqrt{5}}{2} \right)^{m-1} \left(\frac{1-\sqrt{5}}{2} \right)^2 \right\}$$

$$\Rightarrow U_{m+1} = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^{m+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{m+1} \right\}$$

$\therefore P(m+1)$ is true.

Thus, $P(n)$ is true for all $n \leq m \Rightarrow P(n)$ is true for all $n \leq m+1$.

Hence, $P(n)$ is true for all $n \in N$.

EXERCISE 12.2

LEVEL-1

Prove the following by the principle of mathematical induction: (1-38)

1. $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ i.e., the sum of the first n natural numbers is $\frac{n(n+1)}{2}$.

2. $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

3. $1 + 3 + 3^2 + \dots + 3^{n-1} = \frac{3^n - 1}{2}$

4. $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$

5. $1 + 3 + 5 + \dots + (2n-1) = n^2$ i.e., the sum of first n odd natural numbers is n^2 .

6. $\frac{1}{2.5} + \frac{1}{5.8} + \frac{1}{8.11} + \dots + \frac{1}{(3n-1)(3n+2)} = \frac{n}{6n+4}$

7. $\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$

8. $\frac{1}{3.5} + \frac{1}{5.7} + \frac{1}{7.9} + \dots + \frac{1}{(2n+1)(2n+3)} = \frac{n}{3(2n+3)}$

9. $\frac{1}{3.7} + \frac{1}{7.11} + \frac{1}{11.15} + \dots + \frac{1}{(4n-1)(4n+3)} = \frac{n}{3(4n+3)}$

10. $1.2 + 2.2^2 + 3.2^3 + \dots + n.2^n = (n-1)2^{n+1} + 2$

11. $2 + 5 + 8 + 11 + \dots + (3n-1) = \frac{1}{2}n(3n+1)$

12. $1.3 + 2.4 + 3.5 + \dots + n.(n+2) = \frac{1}{6}n(n+1)(2n+7)$

13. $1.3 + 3.5 + 5.7 + \dots + (2n-1)(2n+1) = \frac{n(4n^2 + 6n - 1)}{3}$

14. $1.2 + 2.3 + 3.4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$

15. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} = 1 - \frac{1}{2^n}$

16. $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(4n^2 - 1)$

17. $a + ar + ar^2 + \dots + ar^{n-1} = a \left(\frac{r^n - 1}{r - 1} \right), r \neq 1$

18. $a + (a + d) + (a + 2d) + \dots + (a + (n-1)d) = \frac{n}{2} [2a + (n-1)d]$

19. $5^{2n} - 1$ is divisible by 24 for all $n \in N$

20. $3^{2n} + 7$ is divisible by 8 for all $n \in N$

21. $5^{2n+2} - 24n - 25$ is divisible by 576 for all $n \in N$

22. $3^{2n+2} - 8n - 9$ is divisible by 8 for all $n \in N$

23. $(ab)^n = a^n b^n$ for all $n \in N$

24. $n(n+1)(n+5)$ is a multiple of 3 for all $n \in N$

25. $7^{2n} + 2^{3n-3} \cdot 3^{n-1}$ is divisible by 25 for all $n \in N$

26. $2 \cdot 7^n + 3 \cdot 5^n - 5$ is divisible by 24 for all $n \in N$

27. $11^{n+2} + 12^{2n+1}$ is divisible by 133 for all $n \in N$

28. $1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$ for all $N \in N$.

[NCERT EXEMPLAR]

29. $n^3 - 7n + 3$ is divisible by 3 for all $n \in N$.

30. $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all $n \in N$.

LEVEL-2

31. Prove that $7 + 77 + 777 + \dots + 777 \dots 7 = \frac{7}{81} (10^{n+1} - 9n - 10)$ for all $n \in N$
 $n\text{-digits}$

32. Prove that $\frac{n^7}{7} + \frac{n^5}{5} + \frac{n^3}{3} + \frac{n^2}{2} - \frac{37}{210}n$ is a positive integer for all $n \in N$

33. Prove that $\frac{n^{11}}{11} + \frac{n^5}{5} + \frac{n^3}{3} + \frac{62}{165}n$ is a positive integer for all $n \in N$.

34. Prove that $\frac{1}{2} \tan\left(\frac{x}{2}\right) + \frac{1}{4} \tan\left(\frac{x}{4}\right) + \dots + \frac{1}{2^n} \tan\left(\frac{x}{2^n}\right) = \frac{1}{2^n} \cot\left(\frac{x}{2^n}\right) - \cot x$ for all $n \in N$ and $0 < x < \frac{\pi}{2}$

35. Prove that $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$ for all natural numbers, $n \geq 2$.

36. Prove that $\frac{(2n)!}{2^{2n} (n!)^2} \leq \frac{1}{\sqrt{3n+1}}$ for all $n \in N$.

37. Prove that $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}$ for all $n > 2, n \in N$.

38. Prove that $x^{2n-1} + y^{2n-1}$ is divisible by $x + y$ for all $n \in N$.

39. Prove that $\sin x + \sin 3x + \dots + \sin (2n-1)x = \frac{\sin^2 nx}{\sin x}$ for all $n \in N$.

40. Prove that $\cos \alpha + \cos (\alpha + \beta) + \cos (\alpha + 2\beta) + \dots + \cos (\alpha + (n-1)\beta)$

$$= \frac{\cos \left\{ \alpha + \left(\frac{n-1}{2} \right) \beta \right\} \sin \left(\frac{n\beta}{2} \right)}{\sin \frac{\beta}{2}}$$

for all $n \in \mathbb{N}$.

[NCERT EXEMPLAR]

41. Prove that $\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} > \frac{13}{24}$, for all natural numbers $n > 1$.

[NCERT EXEMPLAR]

42. Given $a_1 = \frac{1}{2} \left(a_0 + \frac{A}{a_0} \right)$, $a_2 = \frac{1}{2} \left(a_1 + \frac{A}{a_1} \right)$ and $a_{n+1} = \frac{1}{2} \left(a_n + \frac{A}{a_n} \right)$ for $n \geq 2$,

where $a > 0$, $A > 0$.

Prove that $\frac{a_n - \sqrt{A}}{a_n + \sqrt{A}} = \left(\frac{a_1 - \sqrt{A}}{a_1 + \sqrt{A}} \right) 2^{n-1}$.

43. Let $P(n)$ be the statement: $2^n \geq 3n$. If $P(r)$ is true, show that $P(r+1)$ is true. Do you conclude that $P(n)$ is true for all $n \in \mathbb{N}$?

44. Show by the Principle of Mathematical induction that the sum S_n of the n terms of the series $1^2 + 2 \times 2^2 + 3^2 + 2 \times 4^2 + 5^2 + 2 \times 6^2 + 7^2 + \dots$ is given by

$$S_n = \begin{cases} \frac{n(n+1)^2}{2}, & \text{if } n \text{ is even} \\ \frac{n^2(n+1)}{2}, & \text{if } n \text{ is odd} \end{cases}$$

[NCERT EXEMPLAR]

45. Prove that the number of subsets of a set containing n distinct elements is 2^n for all $n \in \mathbb{N}$.

[NCERT EXEMPLAR]

46. A sequence $a_1, a_2, a_3, \dots$ is defined by letting $a_1 = 3$ and $a_k = 7a_{k-1}$ for all natural numbers $k \geq 2$. Show that $a_n = 3 \cdot 7^{n-1}$ for all $n \in \mathbb{N}$.

[NCERT EXEMPLAR]

47. A sequence $x_1, x_2, x_3, \dots$ is defined by letting $x_1 = 2$ and $x_k = \frac{x_{k-1}}{n}$ for all natural numbers

$k, k \geq 2$. Show that $x_n = \frac{2}{n!}$ for all $n \in \mathbb{N}$.

[NCERT EXEMPLAR]

48. A sequence $x_0, x_1, x_2, x_3, \dots$ is defined by letting $x_0 = 5$ and $x_k = 4 + x_{k-1}$ for all natural number k . Show that $x_n = 5 + 4n$ for all $n \in \mathbb{N}$ using mathematical induction.

[NCERT EXEMPLAR]

49. Using principle of mathematical induction prove that

$$\sqrt{n} < \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} \text{ for all natural numbers } n \geq 2.$$

[NCERT EXEMPLAR]

50. The distributive law from algebra states that for all real numbers c, a_1 and a_2 , we have

$$c(a_1 + a_2) = ca_1 + ca_2$$

Use this law and mathematical induction to prove that, for all natural numbers, $n \geq 2$, if $c, a_1, a_2, \dots, a_n$ are any real numbers, then

$$c(a_1 + a_2 + \dots + a_n) = ca_1 + ca_2 + \dots + ca_n.$$

VERY SHORT ANSWER TYPE QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per the requirement of the question.

1. State the first principle of mathematical induction.
2. Write the set of value of n for which the statement $P(n): 2n < n!$ is true.
3. State the second principle of mathematical induction.
4. If $P(n): 2 \times 4^{2n+1} + 3^{3n+1}$ is divisible by λ for all $n \in N$ is true, then find the value of λ .

ANSWERS

2. $\{n \in N : n \geq 4\}$ 4. 11

MULTIPLE CHOICES QUESTIONS (MCQS)

Make the correct alternative in each of the following.

1. If $x^n - 1$ is divisible by $x - \lambda$, then the least positive integral value of λ is
(a) 1 (b) 2 (c) 3 (d) 4
2. For all $n \in N$, $3 \times 5^{2n+1} + 2^{3n+1}$ is divisible by
(a) 19 (b) 17 (c) 23 (d) 25
3. If $10^n + 3 \times 4^{n+2} + \lambda$ is divisible by 9 for all $n \in N$, then the least positive integral value of λ is
(a) 5 (b) 3 (c) 7 (d) 1
4. Let $P(n): 2^n < (1 \times 2 \times 3 \times \dots \times n)$. Then the smallest positive integer for which $P(n)$ is true is
(a) 1 (b) 2 (c) 3 (d) 4
5. A student was asked to prove a statement $P(n)$ by induction. He proved $P(k+1)$ is true whenever $P(k)$ is true for all $k > 5 \in N$ and also $P(5)$ is true. On the basis of this he could conclude that $P(n)$ is true.
(a) for all $n \in N$ (b) for all $n > 5$ (c) for all $n \geq 5$ (d) for all $n < 5$
6. If $P(n): 49^n + 16^n + \lambda$ is divisible by 64 for $n \in N$ is true, then the least negative integral value of λ is
(a) -3 (b) -2 (c) -1 (d) -4

ANSWERS

1. (a) 2. (b) 3. (a) 4. (d) 5. (c) 6. (c)

SUMMARY

1. A sentence or description which can be judged to be true or false is called a statement. Statements involving mathematical relations are called mathematical statements.
2. Let $P(n)$ be a statement involving the natural number n such that
(i) $P(1)$ is true.
and, (ii) $P(m+1)$ is true, whenever $P(m)$ is true.
Then, $P(n)$ is true for all $n \in N$.
This is called first principle of mathematical induction.
3. Let $P(n)$ be a statement involving the natural number n such that
(i) $P(1)$ is true
and, (ii) $P(m+1)$ is true, whenever $P(n)$ is true for all $n \leq m$.
Then, $P(n)$ is true for all $n \in N$.
This is called second principle of mathematical induction.

COMPLEX NUMBERS

13.1 NEED FOR COMPLEX NUMBERS

If a, b are natural numbers such that $a > b$, then the equation $x + a = b$ is not solvable in N , the set of natural numbers i.e. there is no natural number satisfying the equation $x + a = b$. So, the set of natural numbers is extended to form the set I of integers in which every equation of the form $x + a = b$; $a, b \in N$ is solvable. But, equations of the form $xa = b$, where $a, b \in I$, $a \neq 0$ are not solvable in I also. Therefore, the set I of integers is extended to obtain the set Q of all rational numbers in which every equation of the form $xa = b$, $a \neq 0$, $a, b \in I$ is uniquely solvable. The equations of the form $x^2 = 2$, $x^2 = 3$ etc. are not solvable in Q because there is no rational number whose square is 2. Such numbers are known as irrational numbers. The set Q of all rational numbers is extended to obtain the set R which includes both rational and irrational numbers. This set is known as the set of real numbers. The equations of the form $x^2 + 1 = 0$, $x^2 + 4 = 0$ etc. are not solvable in R i.e. there is no real number whose square is a negative real number. Euler was the first mathematician to introduce the symbol i (iota) for the square root of -1 i.e. a solution of $x^2 + 1 = 0$ with the property $i^2 = -1$. He also called this symbol as the imaginary unit.

13.2 INTEGRAL POWERS OF IOTA (i)

Positive integral powers of i : We have, $i = \sqrt{-1}$

$$\therefore i^2 = -1, i^3 = i^2 \times i = -i, i^4 = (i^2)^2 = (-1)^2 = 1$$

In order to compute i^n for $n > 4$, we divide n by 4 and obtain the remainder r . Let m be the quotient when n is divided by 4. Then,

$$n = 4m + r, \text{ where } 0 \leq r < 4$$

$$\Rightarrow i^n = i^{4m+r} = (i^4)^m i^r = i^r$$

Thus, the value of i^n for $n > 4$ is i^r , where r is the remainder when n is divided by 4.

Negative integral powers of i : By the law of indices, we have

$$i^{-1} = \frac{1}{i} = \frac{i^3}{i^4} = i^3 = -i, i^{-2} = \frac{1}{i^2} = \frac{1}{-1} = -1$$

$$i^{-3} = \frac{1}{i^3} = \frac{i}{i^4} = i, i^{-4} = \frac{1}{i^4} = \frac{1}{1} = 1$$

If $n > 4$, then $i^{-n} = \frac{1}{i^n} = \frac{1}{i^r}$, where r is the remainder when n is divided by 4

NOTE i^0 is defined as 1.

The above discussion suggests the following algorithm to find integral exponents of i .

ALGORITHM

To find the value of i^n for $n \in \mathbb{Z}$, we may follow the following steps.

STEP I If $n = 0$, then write $i^n = 1$.

STEP II If $n > 0$, then

$$i^n = \begin{cases} i, & \text{if } n = 1 \\ -1, & \text{if } n = 2 \\ -i, & \text{if } n = 3 \\ 1, & \text{if } n = 4 \\ i^r, & \text{if } n > 4, \text{ where } r \text{ is the remainder when } n \text{ is divided by } 4 \end{cases}$$

STEP III If $n < 0$, then $n = -m$, where $m > 0$.

$$\therefore i^n = \begin{cases} i^{-1} = \frac{1}{i} = -i, & \text{if } n = -1 \\ i^{-2} = \frac{1}{i^2} = \frac{1}{-1} = -1, & \text{if } n = -2 \\ i^{-3} = \frac{1}{i^3} = \frac{i}{i^4} = i, & \text{if } n = -3 \\ i^{-4} = \frac{1}{i^4} = \frac{1}{1} = 1, & \text{if } n = -4 \\ i^{-m} = \frac{1}{i^m} = \frac{1}{i^r}, & \text{where } r \text{ is the remainder when } m \text{ is divided by } 4, \text{ if } n < -4. \end{cases}$$

ILLUSTRATIVE EXAMPLES**LEVEL-1**

EXAMPLE 1 Evaluate the following:

(i) i^{135}

(ii) i^{19}

(iii) i^{-999}

(iv) $(-\sqrt{-1})^{4n+3}, n \in \mathbb{N}$

SOLUTION (i) 135 leaves remainder as 3 when it is divided by 4.

$$\therefore i^{135} = i^3 = -i$$

(ii) The remainder is 3 when 19 is divided by 4.

$$\therefore i^{19} = i^3 = -i$$

(iii) We have, $i^{-999} = 1/i^{999}$

On dividing 999 by 4, we obtain 3 as the remainder. Therefore, $i^{999} = i^3$.

$$\text{Hence, } i^{-999} = \frac{1}{i^{999}} = \frac{1}{i^3} = \frac{i}{i^4} = \frac{i}{1} = i$$

(iv) We have, $(-\sqrt{-1})^{4n+3} = (-i)^{4n+3} = (-i)^{4n}(-i)^3 = \{(-i)^4\}^n(-i)^3 = 1 \times -i^3 = i$

EXAMPLE 2 Show that:

(i) $\left\{i^{19} + \left(\frac{1}{i}\right)^{25}\right\}^2 = -4$

(ii) $\left\{i^{17} - \left(\frac{1}{i}\right)^{34}\right\}^2 = 2i$

(iii) $\left\{i^{18} + \left(\frac{1}{i}\right)^{24}\right\}^3 = 0$

(iv) $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$, for all $n \in \mathbb{N}$.

SOLUTION (i) $\left\{i^{19} + \left(\frac{1}{i}\right)^{25}\right\}^2 = \left\{i^{19} + \frac{1}{i^{25}}\right\}^2 = \left\{i^3 + \frac{1}{i}\right\}^2 = \left\{-i + \frac{i^3}{i^4}\right\}^2$

$$= [-i + i^3]^2 = (-i - i)^2 = 4i^2 = -4.$$

$$(ii) \quad \left\{ i^{17} - \left(\frac{1}{i} \right)^{34} \right\}^2 = \left\{ i^{17} - \frac{1}{i^{34}} \right\}^2 = \left\{ i - \frac{1}{i^2} \right\}^2 = \left\{ i - \frac{1}{(-1)} \right\}^2 = (i+1)^2$$

$$= i^2 + 2i + 1 = -1 + 2i + 1 = 2i$$

$$(iii) \quad \left\{ i^{18} + \left(\frac{1}{i} \right)^{24} \right\}^3 = \left\{ i^{18} + \frac{1}{i^{24}} \right\}^3 = \left(i^2 + \frac{1}{1} \right)^3 = (-1+1)^3 = 0$$

$$(iv) \quad i^n + i^{n+1} + i^{n+2} + i^{n+3} = i^n + i^n \times i + i^n \times i^2 + i^n \times i^3$$

$$= i^n (1 + i + i^2 + i^3)$$

$$= i^n (1 + i - 1 - i) = i^n (0) = 0$$

LEVEL-2

EXAMPLE 3 Evaluate $\sum_{n=1}^{13} (i^n + i^{n+1})$, where $n \in N$.

SOLUTION $\sum_{n=1}^{13} (i^n + i^{n+1})$

$$= \sum_{n=1}^{13} (i+1) i^n$$

$$= (i+1) \sum_{n=1}^{13} i^n$$

$$= (i+1) (i + i^2 + i^3 + \dots + i^{13})$$

$$= (i+1) \times i \left(\frac{i^{13} - 1}{i - 1} \right)$$

$$= (i^2 + i) \left(\frac{i - 1}{i - 1} \right)$$

$$= (-1 + i)$$

$$[\because i^{13} = i]$$

EXAMPLE 4 Evaluate $1 + i^2 + i^4 + i^6 + \dots + i^{2n}$.

SOLUTION Let $S = 1 + i^2 + i^4 + i^6 + \dots + i^{2n}$. Then

$$S = 1 + i^2 + (i^2)^2 + (i^2)^3 + \dots + (i^2)^n$$

$$\Rightarrow S = 1 \left\{ \frac{1 - (i^2)^{n+1}}{1 - i^2} \right\} = \frac{1 - (i^2)^{n+1}}{1 + 1} = \frac{1}{2} \{ 1 - (-1)^{n+1} \}$$

$$\Rightarrow S = \begin{cases} \frac{1}{2} (1 - 1) = 0, & \text{if } n \text{ is odd} \\ \frac{1}{2} (1 + 1) = 1, & \text{if } n \text{ is even} \end{cases}$$

EXERCISE 13.1

LEVEL-1

1. Evaluate the following:

(i) i^{457}

(ii) i^{528}

(iii) $\frac{1}{i^{58}}$

(iv) $i^{37} + \frac{1}{i^{67}}$

(v) $\left(i^{41} + \frac{1}{i^{257}} \right)^9$

(vi) $(i^{77} + i^{70} + i^{87} + i^{414})^3$

(vii) $i^{30} + i^{40} + i^{60}$

(viii) $i^{49} + i^{68} + i^{89} + i^{110}$

2. Show that $1 + i^{10} + i^{20} + i^{30}$ is a real number.
3. Find the values of the following expressions:
- (i) $i^{49} + i^{68} + i^{89} + i^{110}$ (ii) $i^{30} + i^{80} + i^{120}$
- (iii) $i + i^2 + i^3 + i^4$ (iv) $i^5 + i^{10} + i^{15}$
- (v) $\frac{i^{592} + i^{590} + i^{588} + i^{586} + i^{584}}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}}$ (vi) $1 + i^2 + i^4 + i^6 + i^8 + \dots + i^{20}$
- (vii) $(1+i)^6 + (1-i)^3$

ANSWERS

1. (i) i (ii) 1 (iii) -1 (iv) $2i$ (v) 0 (vi) -8 (vii) 1 (viii) $2i$
3. (i) $2i$ (ii) 1 (iii) 0 (iv) -1 (v) -1 (vi) 1 (vii) $-2 - 10i$

13.3 IMAGINARY QUANTITIES

The square root of a negative real number is called an imaginary quantity or an imaginary number.

For example, $\sqrt{-3}$, $\sqrt{-4}$, $\sqrt{-9/4}$ etc. are imaginary quantities.

THEOREM If a, b are positive real numbers, then $\sqrt{-a} \times \sqrt{-b} = -\sqrt{ab}$.

PROOF We have,

$$\sqrt{-a} = \sqrt{-1 \times a} = \sqrt{-1} \times \sqrt{a} = i\sqrt{a} \text{ and, } \sqrt{-b} = \sqrt{-1 \times b} = \sqrt{-1} \times \sqrt{b} = i\sqrt{b}$$

$$\therefore \sqrt{-a} \times \sqrt{-b} = (i\sqrt{a})(i\sqrt{b}) = i^2(\sqrt{a} \times \sqrt{b}) = -1(\sqrt{ab}) = -\sqrt{ab}$$

NOTE 1 For any two real numbers $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ is true only when at least one of a and b is either positive or zero. In other words, $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ is not valid if a and b both are negative.

NOTE 2 For any positive real number a , we have $\sqrt{-a} = \sqrt{-1 \times a} = \sqrt{-1} \times \sqrt{a} = i\sqrt{a}$.

ILLUSTRATION 1 Compute the following:

(i) $\sqrt{-144}$

(ii) $\sqrt{-4} \times \sqrt{\frac{-9}{4}}$

(iii) $\sqrt{-25} + 3\sqrt{-4} + 2\sqrt{-9}$

SOLUTION

(i) $\sqrt{-144} = \sqrt{-1 \times 144} = \sqrt{-1} \times \sqrt{144} = 12i$

(ii) $\sqrt{-4} \times \sqrt{\frac{-9}{4}} = (2i) \left(\frac{3i}{2} \right) = 3i^2 = -3$

(iii) $\sqrt{-25} + 3\sqrt{-4} + 2\sqrt{-9} = 5i + 6i + 6i = 17i$

ILLUSTRATION 2 A student writes the formula $\sqrt{ab} = \sqrt{a} \sqrt{b}$. Then he substitutes $a = -1$ and $b = -1$ and finds $1 = -1$. Explain where is he wrong?

SOLUTION Since a and b both are negative. Therefore, $\sqrt{ab}$ cannot be written as $\sqrt{a} \sqrt{b}$. In fact, for a and b both negative, we have $\sqrt{a} \sqrt{b} = -\sqrt{ab}$.

ILLUSTRATION 3 Is the following computation correct? If not give the correct computation:

$$[\sqrt{-2} \cdot \sqrt{-3}] = \sqrt{(-2) \cdot (-3)} = \sqrt{6}$$

SOLUTION The said computation is not correct, because -2 and -3 both are negative and $\sqrt{ab} = \sqrt{a} \sqrt{b}$ is true when at least one of a and b is positive or zero. The correct computation is

$$(\sqrt{-2})(\sqrt{-3}) = (i\sqrt{2})(i\sqrt{3}) = i^2\sqrt{6} = -\sqrt{6}$$